AN INFINITE BASE FIELD SUPPLIES A SEPARABLE CLOSED POINT OF DEGREE AT MOST D ON THE SMOOTH LOCUS OF A PLANE CURVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the chart reduction with the nonzero partial derivative, the etale projection, the fibre-degree bound, and the closing separable-point construction are all retained verbatim.

Editorial note. The label lemma-smooth-plane-curve-point-over-separable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The plane-curve setup and the affine chart. Same chapter, Lemma lemma-equation-plane-curve; Algebra, Section algebra-section-smooth. Identifies the chart D_+(T_0) with the spectrum of k[x, y]/(f) and, via the Jacobian discussion, makes one partial derivative nonzero at the given smooth point.
- Etale projections and universally bounded fibres. Morphisms, Section morphisms-section-universally-bounded. Gives the etale projection V -> Spec(k[x]) after shrinking, the openness of its image, and the bound of degree at most d on the fibres since f has degree at most d in y.

Source reference. Original section 00T1: sources/stacks-chow/tags.txt, source lines 1059–1059. Prerequisite pointer invoked at proof line(s) 1967; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0BYB: sources/stacks-curves/pinned-original-source.tex, source lines 1859–1859. Prerequisite pointer invoked at proof line(s) 1951; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 03J3: sources/stacks-chow/tags.txt, source lines 4625–4625. Prerequisite pointer invoked at proof line(s) 1978; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Z \subset \mathbf{P}^2_k$ be as in Lemma 0BYB and let $I(Z) = (F)$ for some $F \in k[T_0, T_1, T_2]$. If $Z \to \mathrm{Spec}(k)$ is smooth in at least one point and k is infinite, then there exists a closed point $z \in Z$ contained in the smooth locus such that $\kappa(z)/k$ is finite separable of degree at most d.*

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Proof. Suppose that $z' \in Z$ is a point where $Z \to \mathrm{Spec}(k)$ is smooth. After renumbering the coordinates if necessary we may assume z' is contained in $D_+(T_0)$. Set $f = F(1, x, y) \in k[x, y]$. Then $Z \cap D_+(X_0)$ is isomorphic to the spectrum of $k[x, y]/(f)$. Let f_x, f_y be the partial derivatives of f with respect to x, y. Since z' is a smooth point of Z/k we see that either f_x or f_y is nonzero in z' (see discussion in Algebra, Section 00T1). After renumbering the coordinates we may assume f_y is not zero at z'. Hence there is a nonempty open subscheme $V \subset Z \cap D_+(X_0)$ such that the projection

$$p : V \longrightarrow \mathrm{Spec}(k[x])$$

is étale. Because the degree of f as a polynomial in y is at most d, we see that the degrees of the fibres of the projection p are at most d (see discussion in Morphisms, Section 03J3). Moreover, as p is étale the image of p is an open $U \subset \mathrm{Spec}(k[x])$. Finally, since k is infinite, the set of k-rational points $U(k)$ of U is infinite, in particular not empty. Pick any $t \in U(k)$ and let $z \in V$ be a point mapping to t. Then z works. □